

Semen quality of male idiopathic infertile smokers and nonsmokers: an ultrastructural study.

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This retrospective study was aimed at evaluating the effects of cigarette consumption on semen parameters in a group of men with idiopathic infertility. The semen quality of 2 groups of men with idiopathic infertility, smokers (n = 118) and nonsmokers (n = 153), were compared. Conventional semen analysis was performed and sperm morphology was assessed by transmission electron microscopy (TEM). TEM data were elaborated by means of a mathematical formula based on a Bayesian technique able to furnish a fertility index (FI), and the percentages of sperm apoptosis, necrosis, and immaturity. Values of normality recommended by World Health Organization guidelines were used as a control for conventional semen analysis, and values from sperm of 25 men of proven fertility were used for TEM indices. Infertile smoker and nonsmoker patients showed similar sperm parameters, although sperm motility and TEM analysis values in both groups were significantly impaired compared with controls. Smoker patients were then classified as mild (≥ 1 and ≤ 10 cigarettes/d), moderate (> 10 and < 20 cigarettes/day), or heavy smokers (≥ 20 cigarettes/d). Sperm concentration and FI were significantly ($P < .05$) different among the 3 considered smoker classes. Comparing the pairs of smoker classes, sperm concentration and FI in heavy smokers were significantly lower ($P < .05$) than that observed in mild smoker and nonsmoker groups. Although semen quality in males with idiopathic infertility seems not to be dramatically affected by cigarette consumption, heavy smokers show significantly lower sperm concentration and FI: another strong reason to stop smoking.